

Missing Value Analysis Report

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Introduction

This report presents a comprehensive missing value analysis performed on the healthcare datasets used in the AI-Based Personalized Diet and Recovery Planner project. Missing value handling is a critical preprocessing task in healthcare analytics because incomplete data can negatively impact model accuracy, statistical validity, and downstream machine learning workflows.

Objectives

The primary objectives of the missing value analysis were to identify incomplete fields, evaluate missingness patterns, reduce information loss, preserve realistic healthcare relationships, and prepare the datasets for feature engineering and predictive analytics.

Datasets Included

The performed the analysis using the two datasets:

1. cleaned_patients.csv – containing demographic, clinical, and lifestyle information.
2. cleaned_daily_tracking.csv – containing patient monitoring, nutrition, symptom, sleep, and recovery tracking information.

Missing Value Detection Process

The datasets were inspected using pandas profiling techniques including `isnull()`, `sum()`, `info()`, and descriptive statistics. Columns with missing values were categorized into numerical, categorical, optional clinical, and temporal features.

Categorical Missing Values

Categorical fields such as allergies, exercise_done, food_restrictions, and doctor_notes contained missing or blank values. These were handled using domain-appropriate placeholders such as 'None', 'Not Reported', or condition-specific defaults to preserve dataset consistency.

Numerical Missing Values

Numerical features including calorie intake, weight measurements, protein intake, and sleep hours were analyzed for unrealistic null patterns. Median imputation was preferred over mean imputation because healthcare data may contain outliers.

Clinical Notes Handling

Doctor notes and patient observation fields were treated separately because healthcare systems often allow optional narrative entries. Missing notes were either preserved intentionally or generated using synthetic clinical note generation logic.

Impact on Machine Learning

Handling missing values improved dataset completeness, reduced model bias, improved statistical consistency, and ensured compatibility with machine learning algorithms requiring structured numerical input.

Final Outcome

The missing value preprocessing phase successfully minimized incomplete records while maintaining realistic patient relationships and clinical consistency. The cleaned datasets are now suitable for EDA, feature engineering, machine learning, and dashboard analytics.